CHENJIE YANG

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GitHub Hugging Face Homepage

EDUCATION

Southwest Jiaotong University | Junior (3rd-year) | Ph.D. Applicant (Fall 2027)

- GPA: **3.95/4.0** | Rank: **1/220**
- Standardized Tests: TOEFL iBT: 100 | CET-6: **643** | CET-4: 595
- **National Scholarship**, 2024, 2025
- **School-level Annual Outstanding Student Role Model**

RESEARCH INTERESTS

- Embodied Intelligence, World Models, Generative Models.

PUBLICATIONS

Under Review / Preprint

- (Co-First author) *"Automated Manipulation Learning"* | ICML 2026 | Under Review (**6/4/3/4**) | Corr.: Qingyao Wu, SCUT | Project
- (Co-First author) *"RoboRouter: Training-Free Policy Routing for Robotic Manipulation"* | arXiv | Preprint | Corr.: Qingyao Wu, SCUT | Project
- (Co-First author) *"UniTraj: Unified Trajectory-Guided Video Generation"* | IJCAI 2026 | Under Review | Corr.: Yuan Liu, CUHK
- (First author) *"Skill-Conditioned Visual Geolocation for Vision-Language Models"* | ACM MM 2026 | Under Review | Corr.: Chenyu Wu, SWJTU
- (First author) *"ActivityEditor: Learning to Synthesize Physically Valid Human Mobility"* | ACL Rolling Review 2026 | Under Review | Corr.: Junbo Zhang, JD iCity

Accepted

- (Second author; advisor as first) *"Air-Ground Multimodal Transport Planning for Joint Passenger Mobility and Parcel Delivery: Integration of Drones, Aircraft, and Ground Vehicles"* | Transportation Research Part E | Accepted | Corr.: Haoning Xi, The University of Newcastle
- (First author) *"SARIMA-SVR Framework for Great Lakes Water Level Regulation"* | ICML 2025 Workshop NewInML | Accepted | Corr.: Chenjie Yang, SWJTU
- (Co-author) *"A Review of Optimization Methods for Low-Altitude Operations"* | Journal of Transportation Engineering and Information | Accepted | Corr.: Xiaobo Liu, SWJTU

Patent & Software Copyright

- Second inventor (advisor as first), *"A Multimodal Transport Planning Method for Low-Altitude and Ground Traffic Collaboration,"* pending.
- *"Cloud-based ShuMiBee Online Honey Sales Channel Management System,"* Software Copyright.

RESEARCH EXPERIENCE

RoboRouter and Automated Manipulation Learning

South China University of Technology | Project Member Aug 2025–Feb 2026

Instructor: Prof. Qingyao Wu, Associate Professor @ South China University of Technology

- **RoboRouter**: training-free policy routing for language-conditioned manipulation, improving success rates by **3%+** in simulation and **13%+** on real robots, with **79.9%** average simulation success rate; **My contribution**: built the routing pipeline for heterogeneous policies (VLA/VA), fine-tuned **pi0** and **ACT** on **RoboTwin** with **LoRA**, and released checkpoints on HuggingFace and code on GitHub.
- **AutoManipLab**: end-to-end robot policy automation, achieving **0.850** completion score and **2.010** deliverability score on **100** tasks, with **Completion@k / Deliver@k**

of **81% / 51%**; **My contribution**: developed a long-horizon **RAG**-based workflow for task decomposition and tool use, implemented **custom task and randomization**, ManiSkill-ACT skill retrieval, and memory interaction design, and fixed rendering and state/image mismatch issues. Paper submitted to **ICML 2026** (GitHub).

Generative Modeling and Style Alignment

Southwest Jiaotong University | **Project Leader** May 2025–Dec 2025

Instructor: Dr. Junkun Yuan, Research Scientist @ Tencent Hunyuan

- **BAGEL-based style transfer alignment**: preference alignment for style transfer, with a **560-image** benchmark and a **Qwen3-VL** style recognizer achieving **95%+** accuracy; **My contribution**: automated evaluation with **VLM**, and content loss, and ran **DPO** training with **LoRA** on **2 L40 GPUs**. Paper submitted to **ACMMM 2026**.
- **UniTrajectory**: unified trajectory-guided video generation, reducing rotation/translation errors to **1.82 / 5.37** and improving frame consistency to **0.9864**; **My contribution**: contributed to pipeline design, baseline experiments, and manuscript writing. Paper submitted to **IJCAI 2026**.

Video-Model-Based Planning for Autonomous Driving

University of Cambridge | **Summer Program** | **Project Member** Jun 2024–Aug 2024

Instructor: Prof. Pietro Liò, Full Professor @ University of Cambridge

- explored **preference-aligned planning** by mapping **natural-language driving preferences** into structured signals and rolling out future trajectories with **world/video models** under safety constraints; **My contribution**: set up **YOLOv10** and multimodal LLM pipelines, conducted **traffic-sign annotation**, and studied **end-to-end video generation** for planning and embodied transfer.

RL-based Multi-Agent Negotiation for Activity Chain Generation

Tsinghua University | **Intern** | **Project Leader** Aug 2025–Mar 2026

Instructor: Dr. Junbo Zhang, Senior Researcher @ JD iCity

- **ActivityEditor**: dual-agent activity generation, achieving SOTA across four states and improving Arizona **Acc** to **0.780** with **Mint / Atype** reduced to **0.251 / 0.089**; **My contribution**: distilled LLM revision traces and trained **Qwen-Editor** with **SFT** and **GRPO**. Paper submitted to **ACL Rolling Review 2026** (GitHub).
- **HoMe-LLM**: household-aware negotiation for zero-shot multi-person schedule generation, improving accuracy by nearly **20%** on **NHTS 2017**; **My contribution**: implemented the framework with structured prompting and **LoRA**.

Demand Forecasting and Multimodal Planning

Southwest Jiaotong University | **SRTP** | **Project Leader** May 2024–Apr 2025

Instructor: Dr. Haoning Xi, Senior Lecturer @ The University of Newcastle

- **Bike-sharing Rebalancing**: bike-sharing rebalancing via **LSTM**-based demand forecasting and optimization; **My contribution**: developed the forecasting and optimization module.
- **Collaborative Air–Ground Path Planning**: coupled routing for low-altitude and ground vehicles with **MILP + ALNS**; **My contribution**: led modeling, experiments, and writing. Paper accepted by **Transportation Research Part E**.

AWARDS

- **National Champion**, RAICOM Robot Developer Competition National Finals, 2025
- **National First Prize**, Electrician Mathematical Contest in Modeling (EMCM), 2024
- **Provincial First Prize**, Chinese Mathematics Competitions for College Students, 2024

SKILLS

- Technical Stack: **Programming**: Python, C++, C#, SQL | **Methods**: PyTorch, LoRA, SFT, DPO, GRPO, RAG | **Optimization**: MILP, ALNS | **DevOps**: Linux
- Certifications: Career Essentials in Generative AI (**Microsoft & LinkedIn**); NCRE Level 2 (C++); NCRE Level 2 (Python)